

2. Basic functions of the synapses and neurotransmitters (I)

The central nervous system contains more than 100 billion neurons (figure 2). Incoming signals enter the neuron through synapses located mostly on the neuronal dendrites, but also on the cell body. For different types of neurons, there may be only a few hundred or as many as 200,000 such synaptic connections from input fibers. Conversely, the output signal travels by way of a single axon leaving the neuron. Then, this axon has many separate branches to other parts of the nervous system or peripheral body.

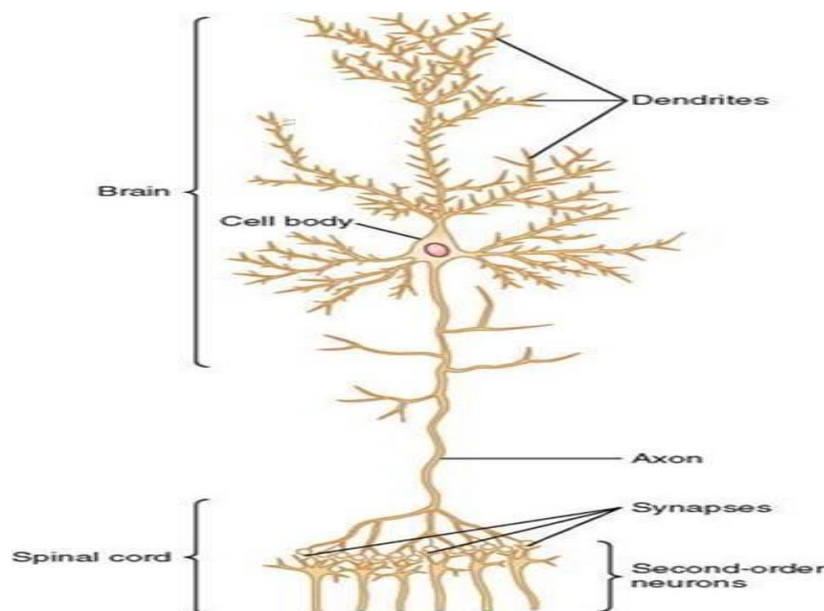


Figure 2: Typical Neuron

A special feature of most synapses is that the signal normally passes only in the forward direction, from the axon of a preceding neuron to dendrites on cell membranes of subsequent neurons. This forces the signal to travel in required directions for performing specific nervous functions.

Sensory Part of the Nervous System—Sensory Receptors (figure 3):

Most activities of the nervous system are initiated by sensory experiences that excite *sensory receptors*, whether visual receptors in the eyes, auditory receptors in the ears, tactile receptors on the surface of the body, or other kinds of receptors. These sensory experiences can either cause immediate reactions from the brain, or memories of the experiences can be stored in the brain for minutes, weeks, or years and determine bodily reactions at some future date.

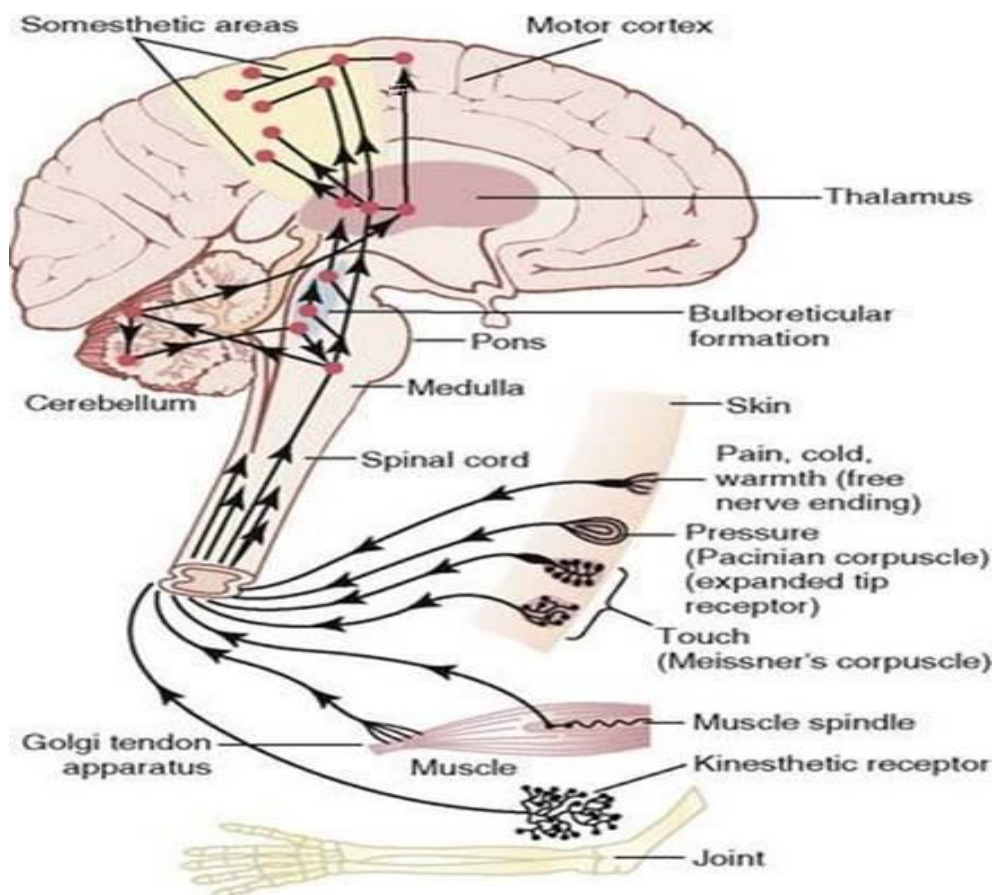


Figure 3: Sensory part and Receptors

Motor Part of the Nervous System—Effectors:

The most important eventual role of the nervous system is to control the various bodily activities. This is achieved by controlling (1) contraction of appropriate skeletal muscles throughout the body, (2) contraction of smooth muscle in the internal organs, and (3) secretion of active chemical substances by both exocrine and endocrine glands

in many parts of the body. These activities are collectively called *motor* functions of the nervous system, and the muscles and glands are called effectors because they are the actual anatomical structures that perform the functions dictated by the nerve signals.

General design of the nervous system:

The unit function of the nervous system is the reflex arc.

The reflex arc:

- A reflex arc is an involuntary response to a stimulus.
- It is composed of (figure 4) :

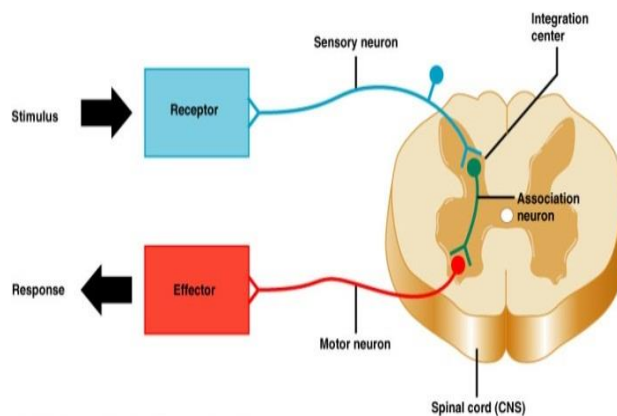


Figure 4: the reflex Arc

1. Receptor.
2. Afferent neuron.
3. Center.
4. Efferent neuron.
5. Effector organ.

- A stimulus: is any change in the environment surrounding a living tissue.

- A receptor: is a specialized structure present at the peripheral end of the afferent nerve fiber and can detect a stimulus. The stimulus acts on the receptor to change the rate of discharge of impulses along the afferent fiber.
- The afferent neuron: conducts impulses from the receptor to the integrating center in the CNS.
- The center: recognizes the stimulus and sends a proper order to peripheral tissues to counteract the environmental changes and produce adjustments (HOMEOSTASIS).
- The efferent neuron: conducts the impulses (orders) from the center to the effector organs.
- The effector organ: produced the response that counteracts the original change in the environment (stimulus).

Types of reflex arc

1. Monosynaptic: if the impulse is relayed from the afferent to the efferent neurone directly (no interneurons in-between) (i.e. there is only one synapse).
2. Polysynaptic: if the impulse is relayed from the afferent to the efferent neuron through ne or more intercommunicating neurons (i.e. there is more than one synapse).
3. Autonomic: when the efferent neuron is a preganglionic nerve fiber that relays in an autonomic ganglion before reaching the effector organ, the reflex is called autonomic reflex. It may be mono- or polysynaptic.

SYNAPSE

Definition: it is the site of contact and not continuity between the axon terminal of one neuron called presynaptic neuron and another neuron called postsynaptic neuron.

Types of synapses:

- Axodendritic synapse: the least excitable.
- Axosomatic synapse:
- Axoaxonic synapse: the most excitable.

Types of synapses (Figure 5):

- a. Chemical: through the release of chemical transmitters. This is the most common type.
- b. Electrical: through direct propagation of nerve impulse from the presynaptic to the postsynaptic neurons without the release of chemical transmitters. Gap junctions between presynaptic and postsynaptic neurons allow ion exchange and impulse propagation. Conduction is very rapid and is present in certain areas of the brain and cerebellum.

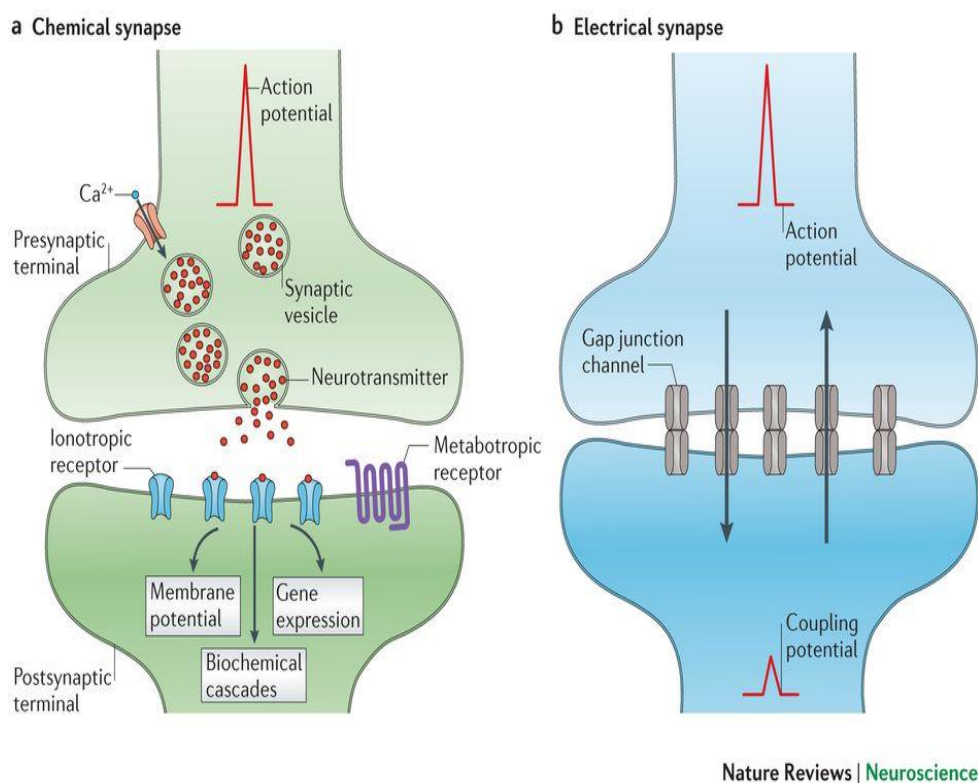


Figure 5 : Types of synapses

Chemical Transmission:

The structure of the synapse (Figure 6):

- The terminal end of the presynaptic neurons breaks into many nerve terminals (synaptic knobs).
- Each synaptic knob is separated from the membrane of the postsynaptic neuron by a space called synaptic cleft containing ECF.
- The synaptic knob contains:
 1. Vesicles containing the chemical transmitter.
 2. Mitochondria to supply energy for synthesis of the chemical transmitter.
- The postsynaptic membrane under the knob contains specific receptors for the chemical transmitter.
- The combination of the chemical transmitter with its receptor changes the permeability of the postsynaptic membrane to ions → change the membrane potential of the postsynaptic neuron and consequently its excitability.

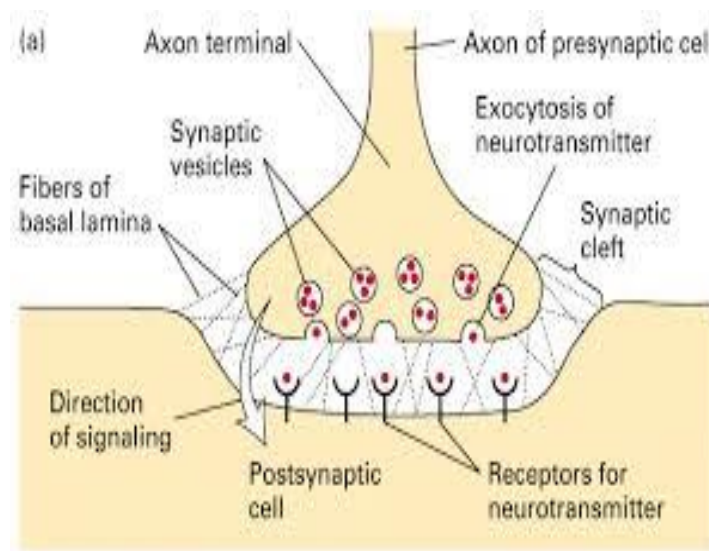
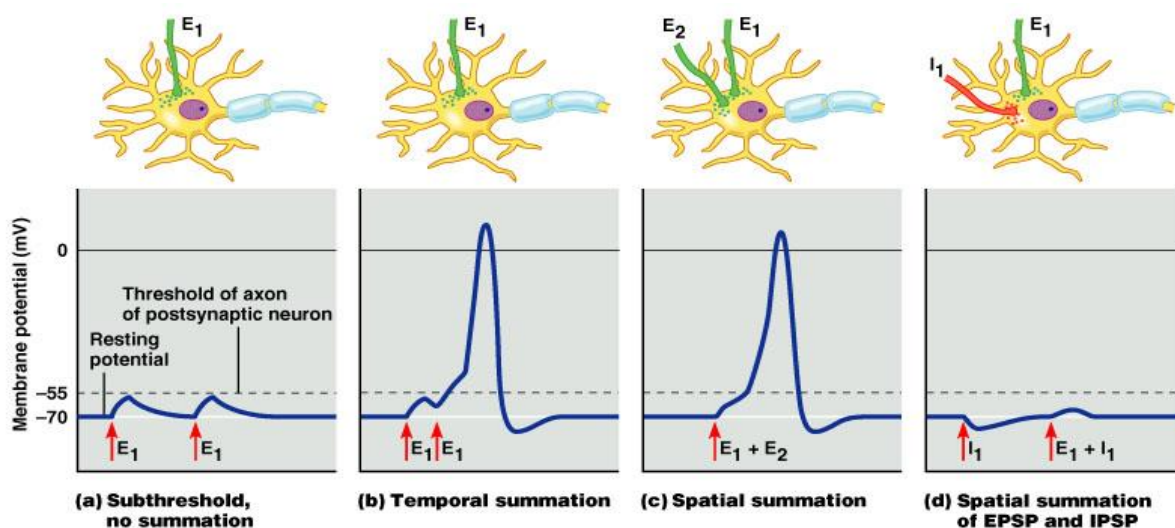


Figure 6: Synaptic Transmission

Mechanism of synaptic transmission (figure 7):

- The membrane of the synaptic knob as well as the postsynaptic neuron under resting conditions is in the polarized state (RMP).
- The presynaptic nerve impulse (1) reaches the terminal knobs as a wave of reversal polarity $\rightarrow$ (2) $\uparrow$ the permeability of to $\text{Ca}^{2+} \rightarrow \uparrow \text{Ca}^{2+}$ diffusion into the terminal knobs $\rightarrow$ (3) stimulates exocytosis of knob's vesicles $\rightarrow$ release of transmitter into the synaptic cleft. (4) the transmitter crosses the cleft and binds to its receptors on the postsynaptic neuron changing the permeability to ions according to the type of synapse:
- Excitatory synapses (EPSP): (5) the excitatory transmitter (e.g. ACh, catecholamines, serotonin,..etc) $\rightarrow$ (6) $\uparrow$ the permeability of postsynaptic membrane to Na^+ and to K^+ but more to Na^+ $\rightarrow \uparrow \text{Na}^+$ influx into postsynaptic neuron $\rightarrow$ (7) depolarization, with increased excitability (EPSP).
- Inhibitory synapse (IPSP): (5) the inhibitory transmitter (e.g. GABA, Glycine, ..etc) binds to its receptors $\rightarrow$ (6) $\uparrow$ permeability of postsynaptic membrane to K^+ and $\text{Cl}^- \rightarrow \uparrow \text{K}^+$ outflux and Cl^- influx into the postsynaptic neurone $\rightarrow$ (7) hyperpolarization, with decreased excitability (IPSP).

EPSP & IPSP



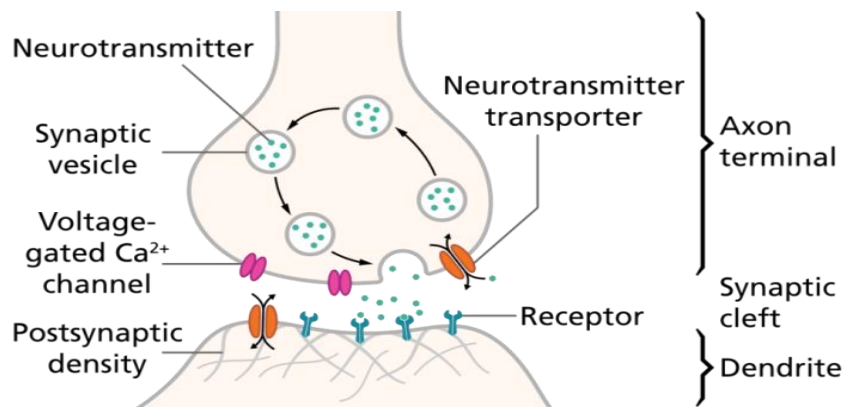


Figure 7 Mechanism of synaptic transmission

Miniature excitatory or inhibitory postsynaptic potential (mEPSP and mIPSP):

It is a process of partial postsynaptic depolarization (mEPSP) or hyperpolarization (mIPSP) produced by spontaneous resting flow of Ca^{2+} into the presynaptic terminal knob with the release of few chemical transmitter molecules independent of any nerve impulse. The potential change is not sufficient to produce any postsynaptic discharge.

Mechanism of AP generation in the postsynaptic neuron:

The postsynaptic neuron is divided into two functional divisions:

- The dendrites, cell body and axon hillock (receptive field): their membrane potential has the same properties as the generator (receptor) potential.
- The axon: initiates and conducts the nerve impulse.

A single presynaptic excitatory impulse is not sufficient to depolarize the postsynaptic neuron to the FL to discharge. The receptor potential created needs summation. One of the following mechanisms occurs (figure 8):

1. Temporal Summation: Repeated stimulation of a single presynaptic fiber at a high rate so that the EPSP of each stimulus occurs before the end of the

preceding one → summation of depolarization occurs bringing the membrane potential to of the postsynaptic neuron to the FL to discharge.

2. Spatial Summation: Simultaneous stimulation of several excitatory presynaptic neurons that converge on the same postsynaptic neuron will lead to summation of EPSP produced by each presynaptic fiber. This brings the membrane potential of the postsynaptic neuron to the FL to discharge.

The AP in the postsynaptic neuron starts usually at the axon hillock because:

- It has low threshold (i.e. more excitable).
- It contains many Na^+ channels.

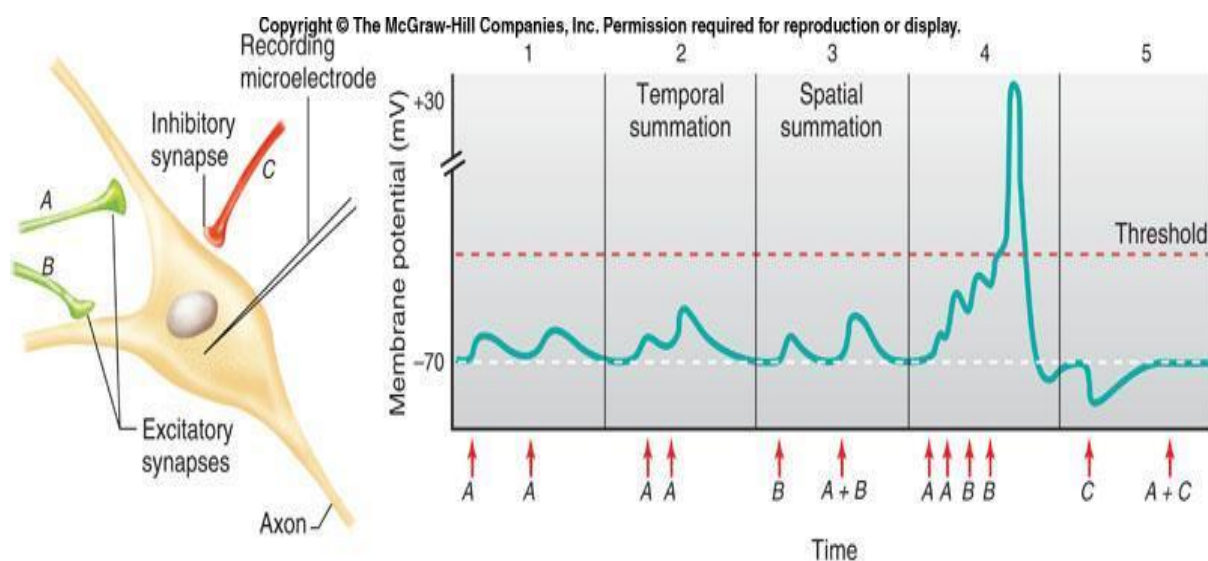


Figure 8: temporal and spatial summation

Types of synaptic inhibition (figure 9):

1. Postsynaptic Inhibition: see IPSP.
2. Presynaptic Inhibition:
 - In this type, inhibitory knobs lie directly on another excitatory presynaptic terminal knob.

- Stimulation of the inhibitory knob → releases inhibitory transmitter (e.g. GABA, Glycine,..etc) which binds to specific receptors on the excitatory presynaptic knob → decreases the amplitude of excitatory presynaptic potential → decrease the permeability of the presynaptic excitatory knob to Ca^{2+} → decreases the release of excitatory transmitter failure to bring the EPSP to the FL to discharge → synaptic inhibition.

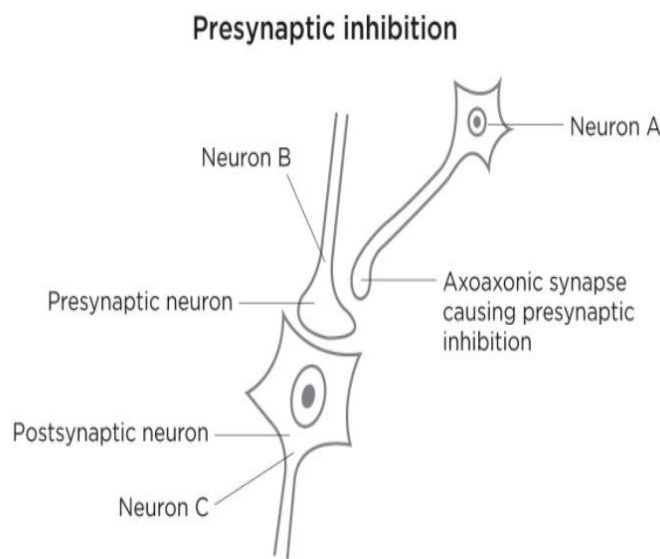


Figure 9: presynaptic inhibition

3. Lateral Inhibition: It is important for certain neurological processes e.g.

A. Reflex responses: contraction of agonists is associated with inhibition of antagonists

B. Localization of sensory stimulus:

Mechanism: stimulation of excitatory presynaptic fibers excites a postsynaptic neuron through the release of excitatory transmitter. At the same time, collateral from the presynaptic fiber excites an inhibitory interneuron that releases an inhibitory

transmitter to inhibit the nearby postsynaptic neuron. The inhibitory interneuron in the spinal cord is called Renshaw cell.

Types Of Synaptic Receptors

1. Presynaptic receptors: they are often inhibitory (α receptors).
2. Postsynaptic receptors:
 1. Cholinergic : 2 types nicotinic (N) and muscharinic (M).
 2. Adrenergic: 2 types α and β .
 3. Dopaminergic: 5 types (D1---D5).
 4. Seretonergeric: 7 tytypes.
 5. Gabergic: 3 types (A and B in CNS and C in the retina).
 6. Histaminergic: 3 types (H1—H3).